Stable Diffusion

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What is Stable Diffusion?

- A state-of-the-art text-to-image generative model.
- Released by Stability AI and collaborators (2022).
- Open-source, enabling wide adoption and customization.

Advantages

- High-quality, diverse image generation.
- Efficient training and inference due to latent space operations.
- Open-source and extensible for research and applications.

What do we need to make a good generative model?

1. Method of learning to generate new images — **Forward/reverse diffusion**
2. Way to link text and images — **Text-image representation model**
3. Way to compress images — **Autoencoder**
4. Way to add in good inductive biases — **U-net architecture + ‘attention’**

Making a ‘good’ generative model is about making all these parts work together well!

Generative Modeling

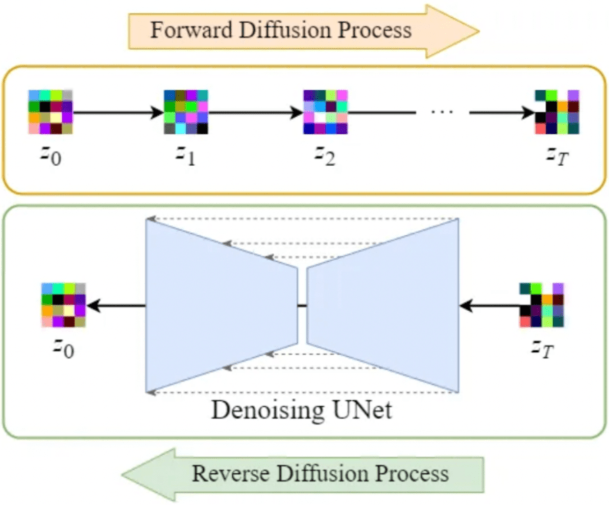
- Early successes: GANs and VAEs enabled impressive image synthesis and data generation.
- Drawbacks:
  - GANs: Training instability, mode collapse, and sensitivity to hyperparameters.
  - VAEs: Blurry outputs and limited expressiveness.
- Key Challenge: Achieving both high sample quality and stable, reliable training.

Where do we go from here?

- Need for models that are robust, interpretable, and capable of generating diverse, high-fidelity samples.

Enter Diffusion Models:

- Inspired by thermodynamics, diffusion models gradually transform noise into data.
- Highlights:
  - Simple training objective
  - Stable optimization
  - State-of-the-art sample quality
  - Probabilistic and interpretable framework


Forward Diffusion Process
z_0
z_1
z_2
z_T
z_0
z_T
Denoising UNet
Reverse Diffusion Process

Quick Overview

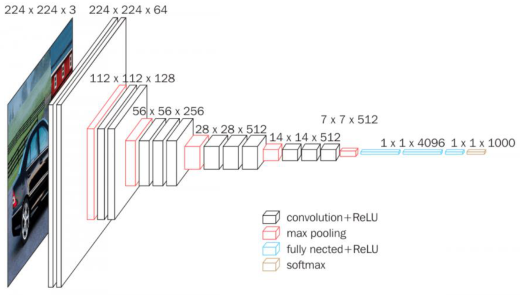
- **Forward:** Clean $\rightarrow$ Noise (additive Gaussian steps)
- **Reverse:** Learn **U-Net** to remove noise step by step
- **Training:** Model predicts the noise added
- **Sampling:** Roll back from pure noise to a data sample

Stable Diffusion Core Backbone:

Attention-U-Net Architecture

Refresher: Convolutional Neural Network

- CNN parametrizes a function over images.
- **Motivation:**
  - Features are translationally invariant.
  - Extract features at different scales / abstraction levels.
- **Key modules:**
  - Convolution
  - Downsampling (max-pool)
- Features of larger scale (larger receptive field) $\Rightarrow$ higher abstraction level.

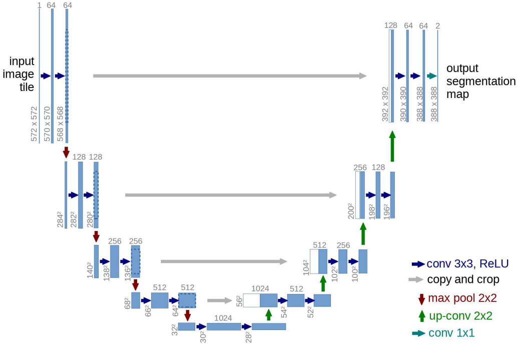


Binxu Wang, Stable Diffusion: A Tutorial (Harvard ML from Scratch).

CNN + inverted CNN ⇒ U-Net

- An inverted CNN (generator) can generate images.
- CNN + inverted CNN can model an image-to-image function.
- **Down sampling:** convolution.
- **Up sampling:** transposed convolution.

U-Net: A Natural Architecture for Image-to-Image Functions



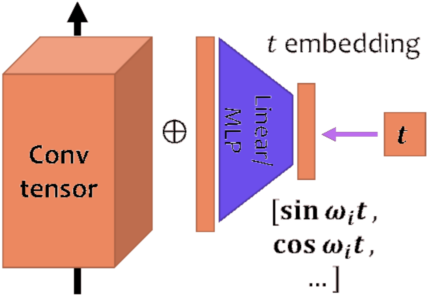
Binxu Wang, Stable Diffusion: A Tutorial (Harvard ML from Scratch).

The score function is time-dependent.

Target: $s(x,t) = \nabla_x \log p(x,t)$

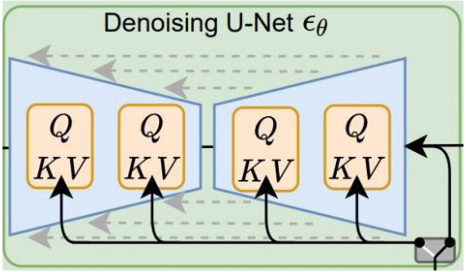
Add time dependency

- Assume time dependency is spatially homogeneous.
- Add one scalar value per channel $f(t)$.
- Parametrize $f(t)$ by MLP/linear of Fourier basis.
  - Fourier features: $[\sin \omega_i t, \cos \omega_i t, \ldots]$



Binxu Wang, Stable Diffusion: A Tutorial (Harvard ML from Scratch).

U-Net in Stable Diffusion

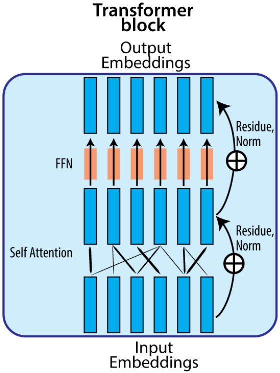


Binxu Wang, Stable Diffusion: A Tutorial (Harvard ML from Scratch).

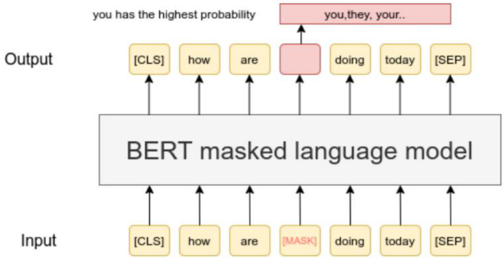
- Input convolution: Projects 4-channel input $\rightarrow$ 320 channels using a 2D convolution.
- Time embedding: inject time information into the network.
- Downsampling: sometimes interleaved with cross-attention.
- Mid block: applies additional residual and (cross-)attention operations.
- Upsampling, normalization, activation, output convolution.

How to Understand Prompts: Language Model Refresher

- Meaning of a word depends on context, not always the same.
- Example: "I *book* a ticket to buy that *book*."
- Word vectors should depend on context.
- Transformers let each word absorb influence from other words to become contextualized.

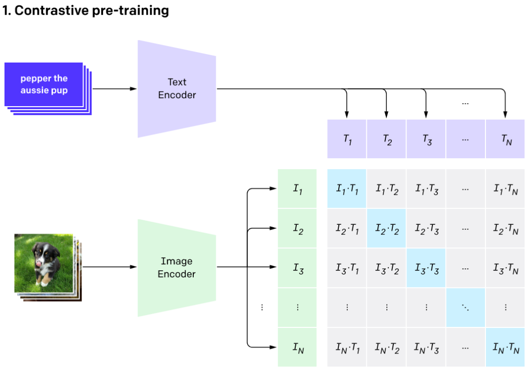


- Self-supervised learning of word representations.
- Predict missing / next words in a sentence (BERT, GPT).
- Contrastive learning: match image and text (CLIP).
- Downstream classifiers can decode part of speech, sentiment, . . .



Joint Representation for Vision and Language: CLIP

- Learn a joint encoding space for text captions and images.
- Maximize representation similarity between an image and its caption.
- Minimize similarity for other pairs.



Adapted from: Radford et al., CLIP, arXiv:2103.00020

Choice of Text Encoding

- Encoder in Stable Diffusion: pre-trained CLIP ViT-L/14 text encoder.
- Word vectors can also be randomly initialized and learned online.
- **Other conditional signals:**
  - Object categories (e.g. Shark, Trout): one vector per class.
  - Face attributes (e.g. {female, blonde hair, with glasses}): one vector per attribute.
- Time to be creative!

Perceptual vs. Semantic Compression

Pixel-space diffusion is redundant: the U-Net spends capacity on imperceptible high-frequency detail. **Stable Diffusion** (Rombach et al., 2022) decouples:

1. **Perceptual compression (VAE):** $\mathcal{E} : \mathbf{x} \mapsto \mathbf{z}$; remove imperceptible spatial detail, preserve layout.
2. **Semantic generation (LDM):** run diffusion on $\mathbf{z}$; lower memory, faster training.
3. **Decode:** $\mathcal{D}(\mathbf{z}) \mapsto \mathbf{x}$; project back to pixels.

Typical spatial downsampling: $8\times$ (e.g., $512 \times 512 \times 3$ pixels $\rightarrow$ $64 \times 64 \times 4$ latents, a $\sim 48\times$ reduction in dimensions).

Cross-Attention: Injecting Text into the U-Net

Pipeline: CLIP text encoder $\rightarrow$ token embeddings $\mathbf{c} \in \mathbb{R}^{L \times d}$.

- **Self-attention:** Q, K, V all come from spatial latent features; provides global visual coherence.
- **Cross-attention:** Q from spatial features; K, V from the text embeddings; aligns image patches with prompt tokens.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

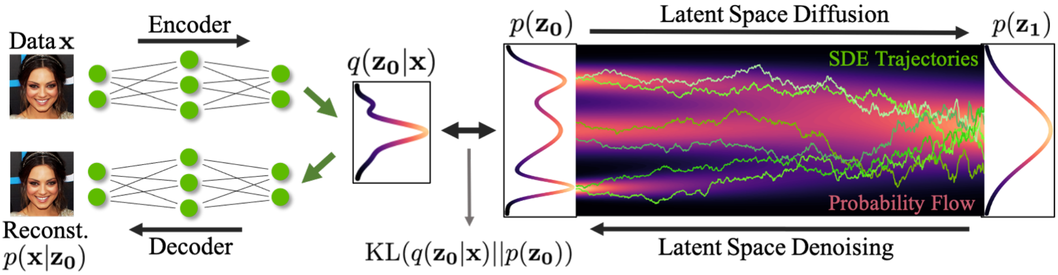
Scaling by $\sqrt{d_k}$ prevents softmax saturation in high dimensions.

Token length $L \leq 77$ (CLIP); multi-head attention throughout the U-Net blocks.

Stable Diffusion:

A Latent Diffusion Model (LDM)

What is a Latent Space?



- A **latent space** is an abstract, compressed space where high-dimensional data (like images) is represented in a lower-dimensional form.
- Think of it as a hidden layer of meaning—each point in this space represents a potential image or concept.
- In classical models like GANs or VAEs, the latent space is directly sampled from (e.g., a vector $\mathbf{z}$), and the generator turns it into an image.

What is a Latent Space? (cont.)

- **Latent** space = "Hidden space of ideas or features"

Where is the Latent Space in Diffusion?

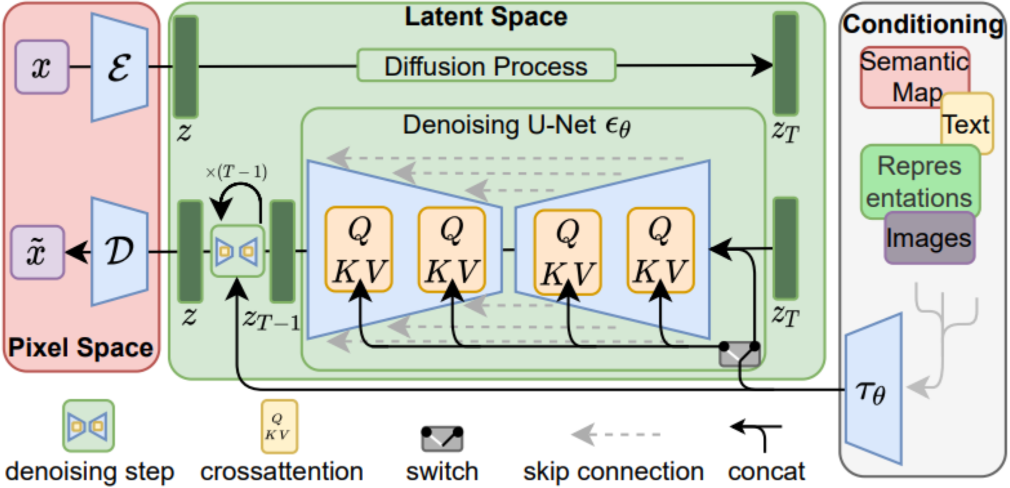
Two main ways latent spaces are used in diffusion:

- **1. Latent Diffusion Models (LDMs):**
  - Diffusion is performed in a compressed latent space, not directly on high-dimensional images.
  - **Pipeline:** Compress image $\rightarrow$ Denoise in latent space $\rightarrow$ Decode back to image.
  - Enables faster, cheaper training and inference, and allows for semantic control (e.g., text prompts, masks).
- **2. Latent Representations in Noise Vectors:**
  - Even in standard diffusion models, the initial noise vector acts as a latent code.
  - Changing the noise vector generates different images or morphs between them.

Why Latent Diffusion?

- **Traditional pixel-space diffusion models are computationally expensive**: they operate directly on high-dimensional images, making both training and inference costly.
- **Latent diffusion reduces computational cost** by performing the diffusion process in a compressed, lower-dimensional latent space—while still maintaining high-quality image generation.
- This efficiency enables more complex and practical applications, such as text-guided generation and flexible image editing.
- **How is this achieved?** Instead of working in pixel space, we learn a compressed latent space using a Variational Autoencoder (VAE) with very low KL divergence weight (e.g., 10^{-6}). This yields a compact representation suitable for diffusion.
- This approach mirrors ideas from autoregressive models (e.g., VQGAN), enabling even more efficient and controllable generation pipelines.

Latent Diffusion Models: Visual Overview



Stable Diffusion:

Guidance and Sampling

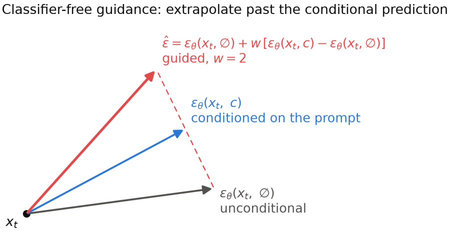
Classifier-Free Guidance (CFG)

- Conditioning alone follows the prompt only loosely; every real system amplifies it with **classifier-free guidance** (Ho & Salimans, 2022).
- **Training trick:** randomly drop the text conditioning for a fraction of training steps ($\sim$10%), so one network learns both a **conditional** and an **unconditional** noise prediction.
- **At sampling time**, run both and **extrapolate**:

$$\hat{\epsilon} = \epsilon_\theta(x_t, \varnothing) + w\left[\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t, \varnothing)\right]$$

- The guidance scale w trades **prompt fidelity** against **diversity**: $w = 1$ is plain conditional sampling; SD v1 defaults to $w \approx 7.5$; too high causes oversaturated, artifact-prone images.

CFG, Geometrically



- Each denoising step pushes the sample **past** the conditional prediction, away from the unconditional one: whatever the prompt adds gets amplified.
- **Negative prompts** reuse the same mechanism: replace the unconditional branch with an embedding of what you do *not* want, and the extrapolation pushes away from it.

Samplers: From 1000 Steps to 20

- DDPM training uses ∼1000 noise levels, but sampling does not have to visit all of them.
- **DDIM** (Song et al., 2021): a deterministic sampler over the same trained model; skips steps and enables consistent latent-space edits. Typically 20–50 steps.
- **ODE-solver view:** sampling is integrating a differential equation, so higher-order solvers (DPM-Solver++, Euler variants in practice) reach good quality in fewer steps.
- **Distillation** goes further, to 1–4 steps: latent consistency models (LCM), adversarial diffusion distillation (SDXL Turbo), and SD 3.5 **Large Turbo** all distill a many-step teacher into a few-step student.
- Practical takeaway: the model, the sampler, and the step count are **independent choices**; most quality-speed tuning happens here.

Customizing Stable Diffusion

Personalization: Textual Inversion, DreamBooth, LoRA

- Open weights made SD the platform people **fine-tune**; three techniques cover most of it:
- **Textual Inversion** (Gal et al., 2023): learn a **new token embedding** for a concept from a few images; the rest of the model stays frozen ($\sim$kilobytes per concept).
- **DreamBooth** (Ruiz et al., 2023): fine-tune the model itself to bind a **rare token** to a specific subject (your pet, a product), with a prior-preservation loss against forgetting the class.
- **LoRA** (Hu et al., 2022): train **low-rank adapters** on the attention weights instead of the full model; megabytes instead of gigabytes, composable, and cheap to share. The de facto standard for style and character adaptation.

Spatial Control: ControlNet

- Text describes *what*; it is poor at *where*. **ControlNet** (Zhang et al., 2023) adds spatial conditioning: edge maps, human pose, depth, sketches, segmentation.
- Mechanism: clone the U-Net encoder into a **trainable copy** that ingests the condition image, keep the original weights **frozen**, and connect the copy through **zero-initialized convolutions** so training starts from a no-op and cannot damage the base model.
- Result: precise layout control with modest paired data, reusing everything the base model knows.
- Together with LoRA and inpainting, this is why an open 2022 model still anchors production pipelines today.

Stable Diffusion:

Applications and Lineage

Application: Stable Video Diffusion

- Initialize from Stable Diffusion.
- Add temporal layers (e.g., 1D Conv, temporal attention).
- Finetune on video data.
- Data pipeline:
  - Scrape **unlabeled** videos from the internet.
  - Use image and video captioners to generate synthetic text labels.
  - Apply aggressive data filtering using several heuristics (CLIP score, optical flow score, aesthetic scores, OCR, etc.).

Application: Stable Video Diffusion (cont.)

"a man wearing a white hat"

Image Inpainting with GLIDE

[0] GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models

a shiba inu wearing a beret and black turtleneck

a close up of a handpalm with leaves growing from it

Text-to-image generation with DALL·E 2

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

Image Variations

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

Interpolate image CLIP embeddings $\mathbf{z}_i$.

Use different $\mathbf{x}_T$ to get different interpolation trajectories.

Image interpolation

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

a photo of a cat → an anime drawing of a super saiyan cat, artstation

a photo of a victorian house → a photo of a modern house

a photo of an adult lion → a photo of lion cub

Change the image CLIP embedding towards the difference of the text CLIP embeddings of two prompts.

Decoder latent is kept as a constant.

Text Difference Image interpolation

[0] Hierarchical Text-Conditional Image Generation with CLIP Latents

Imagen - Google

A brain riding a rocketship heading towards the moon.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Imagen - Google

A dragon fruit wearing karate belt in the snow.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Imagen - Google

A relaxed garlic with a blindfold reading a newspaper while floating in a pool of tomato soup.

[0] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

What changed vs. Stable Diffusion v1?

- **New text encoder:** OpenCLIP (LAION + Stability AI) replaces CLIP ViT-L/14—stronger language understanding and image quality.
- **Native resolutions:** text-to-image at **512**$\times$**512** and **768**$\times$**768**.
- **Upscaler diffusion:** $\times$4 super-resolution (e.g. 128$\rightarrow$512).
- **depth2img:** depth-guided image-to-image—structure-preserving synthesis from inferred depth + text.
- **Updated inpainting:** text-guided inpainting fine-tuned on the Stable Diffusion 2.0 base model.

[0] Lopez, "Stable Diffusion 2.0 Release," Stability AI (Nov 2022).

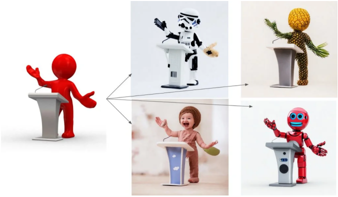
Stable Diffusion 2.0: Upscaler Diffusion

- Dedicated **upscaler diffusion** model enhances resolution by $\times 4$.
- Example: **128**$\times$**128** low-res sample $\rightarrow$ **512**$\times$**512** output.

Left: 128$\times$128. Right: 512$\times$512 upscaled.

[0] Stability AI, Stable Diffusion 2.0 Release.

- **depth2img** infers depth from an input image, then generates new images conditioned on **text** + **depth**.
- Enables structure-preserving image-to-image and shape-conditional synthesis.
- Can produce radically different appearances while preserving spatial coherence and depth.



Input → variants (top); depth coherence preserved (bottom).

[0] Stability AI, Stable Diffusion 2.0 Release.

Stable Diffusion 2.0: Inpainting

Text-guided inpainting model fine-tuned on the SD 2.0 text-to-image base.

[0] Stability AI, Stable Diffusion 2.0 Release.

Application: SDXL

- **SDXL** (Podell et al., 2023): the scaled-up successor to SD 2; for a long time the most-used open text-to-image model.
- A 2.6B-parameter U-Net (about 3$\times$ the 860M of SD 1/2) with **two text encoders** used together (OpenCLIP ViT-bigG + CLIP ViT-L).
- Native **1024**$\times$**1024** generation, plus micro-conditioning on image size and crop so the model learns from mixed-resolution data without artifacts.
- Optional **refiner** model: a second latent-diffusion stage specialized in high-frequency detail.

Inside Stable Diffusion 3: MM-DiT and Rectified Flow

- SD 3 changes both the backbone and the training objective (Esser et al., 2024).
- **Backbone:** the U-Net is replaced by a **multimodal diffusion transformer (MM-DiT)**, following the DiT line (Peebles & Xie, 2023): text and image tokens flow through a joint transformer with separate weights per modality.
- **Objective: rectified flow**, i.e. flow matching on straight noise-data paths, instead of the DDPM objective; the paper's ablations found it superior at scale. (See the flow-matching section of the Normalizing Flows lecture.)
- **Text encoders:** two CLIP encoders plus **T5-XXL** for long, compositional prompts and better in-image text.
- **FLUX.1** (Black Forest Labs, 2024), built by SD's original authors, continues this rectified-flow-transformer lineage as the strongest open release of its generation.

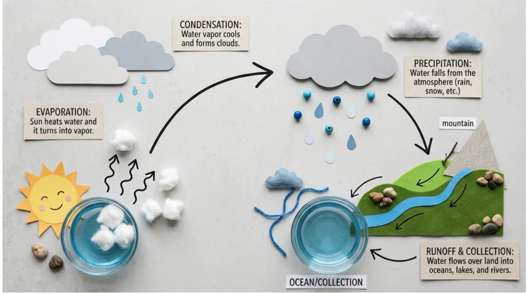
Stable Diffusion 3.5: Diverse & Versatile Outputs

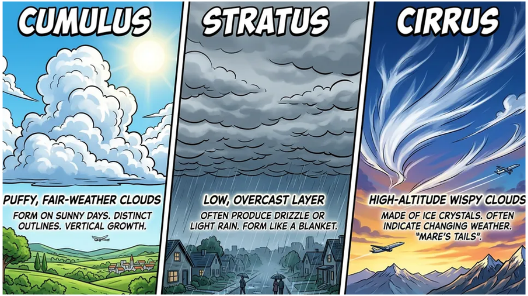
Prompt: photo of three potions: the first potion is blue with the label "MANA", the second potion is red with the label "HEALTH", the third potion is green with the label "POISON". Old apothecary.

Prompt: scene of a giant ancient tortoise with a fantasy city built on its back. The tortoise's shell is covered in lush, dense forest with towering trees and a hidden, misty village nestled in the foliage. The city consists of intricately designed buildings that blend seamlessly with the natural environment, featuring rope bridges connecting different sections of the city.

Prompt: collage art 'We're Leaving For the Future' 1980s #vaporwave aesthetic internet art glued layered magazine cutout image shape scrap, torn ragged paper art, BASIC code, halftone, #pixelart.

- **Customizability:** Large (8.1B), Large Turbo (8.1B) and Medium (2.5B) models.
- **Efficient Performance:** Optimized to run on standard consumer hardware; Medium and Large Turbo require no specialized GPUs.
- **Highly Accessible:** Medium needs only 9.9 GB of VRAM, making advanced diffusion imaging feasible on most consumer GPUs.





Examples: water-cycle infographic and cloud-type comparison.

- Builds on Google's Gemini Image line: Nano Banana (2025) → Nano Banana Pro → Nano Banana 2.
- Production-ready output: multiple aspect ratios and resolutions from 512px to 4K.

[0] Raisinghani, "Nano Banana 2: Combining Pro capabilities with lightning-fast speed," Google DeepMind blog (Feb 2026).

Try It Yourself!

Text to Image generation with stable diffusion.
https://huggingface.co/spaces/stabilityai/stable-diffusion

[0] High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion:

Summary & Limitations

From diffusion to Stable Diffusion

- Diffusion models learn to reverse a gradual noising process; sampling remains iterative but training is stable and scalable.
- **Latent Diffusion (LDM):** denoise in a VAE-compressed latent space ($\sim 8\times$ downsampling) rather than full pixel space—lower memory and faster training/inference.

Core architecture

- **Attention U-Net:** encoder–decoder with skip connections; downsampling / upsampling for multiscale features.
- **Time conditioning:** Fourier-feature time embeddings added per channel.
- **Text conditioning:** CLIP (ViT-L/14 in SD v1) maps prompts to token embeddings; cross-attention injects language into the U-Net.

Using and steering the model

- **Classifier-free guidance:** extrapolate past the conditional prediction with scale w; negative prompts reuse the same mechanism.
- **Samplers:** DDIM and higher-order ODE solvers cut 1000 steps to 20–50; distillation (LCM, Turbo) reaches 1–4.
- **Customization:** Textual Inversion, DreamBooth, and LoRA personalize; ControlNet adds spatial control.

Summary (cont.)

Model evolution & applications

- **SD 2.0:** OpenCLIP text encoder; native 512/768 T2I; upscaler ($\times$4), depth2img, and updated inpainting.
- **SDXL:** 3.5B U-Net, dual text encoders, native 1024^2, optional refiner.
- **SD 3 / 3.5:** MM-DiT transformer backbone trained with rectified flow; Large 8.1B, Large Turbo, Medium 2.5B run on consumer GPUs; FLUX.1 continues the lineage.
- Broader ecosystem: video (Stable Video Diffusion), inpainting/editing (GLIDE, DALL·E 2), and newer multimodal systems (e.g. Nano Banana 2).

Limitations

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- **Slow sampling** — many denoising steps (partially addressed by distilled models such as SD 3.5 Large Turbo).
- **Data & compute** — high-quality open models still require large-scale curated data and substantial GPU resources to train.
- **Prompt sensitivity** — output quality and adherence depend on prompt wording; base models may vary across seeds.
- **Distribution gaps** — rare concepts, fine details, and compositional reasoning remain challenging.
- **Safety & misuse** — text-to-image systems raise concerns around deepfakes, copyright, and misinformation.

Tutorials and Surveys

- Binxu Wang: Stable Diffusion: A Tutorial (Harvard ML from Scratch) — U-Net, time embedding, CLIP conditioning.
- Kreis, K., Gao, R., & Vahdat, A.: CVPR 2022 Tutorial: Diffusion Models
- Rogge, N., & Rasul, K.: The Annotated Diffusion Model
- Lilian Weng: What are Diffusion Models?
- Yang Song: Score-Based Generative Modeling: A Brief Overview

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Stable Diffusion: Core Papers

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- Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP) — text encoder and cross-attention conditioning.

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Applications and Model Releases

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- Ramesh, A., et al. (2022). Hierarchical Text-Conditional Image Generation with CLIP Latents (DALL·E 2)
- Saharia, C., et al. (2022). Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (Imagen)
- Blattmann, A., et al. (2023). Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets
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